

Bharatiya Vidya Bhavan's
SARDAR PATEL INSTITUTE OF TECHNOLOGY
(An Autonomous Institute Affiliated to University of Mumbai)
MUNSHI NAGAR, ANDHERI (WEST), MUMBAI - 400058
MID SEMESTER EXAMINATION

Sr.No. **8505**

AV: 2025-26

Sem: 1

Exam Type: Written

Session: Odd MSE October 2025

Slot: 10:00am - 11:00am

Block: 3D1

Date: 08-10-2025

C Code: CS413

Seat No: 2023601009

1_29_533_986_1_1505_38376

Name of the candidate. Sheeba Suleman Shaikh

Candidate's UID number:

2	0	2	3	6	0	1	0	0	9
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Examination class room number:

4	0	1
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Invigilator's signature with date:

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(To be filled in by the candidate)

EXAMINATION: B.E. BTECH MSE
(For example FY MCA/FY BTECH)

PROGRAM: CSE DITP

DAY AND DATE: wednesday, 08/10/25

COURSE NAME: DL

SEMESTER : 7

I / II / III / IV / V / VI / VII / VIII

(To be filled in by the examiner)								
Question numbers	1	2	3	4	5	6	Total	Signature of the examiner
Marks awarded								
Marks after revaluation								

INSTRUCTIONS TO CANDIDATES

- 1) Candidate should ensure that all answer books including supplementary answers books received by them bear the signature of the junior supervisor, otherwise the answer books will not be examined.
- 2) Begin the answer to each question on a new page. For each answer, write the corresponding question number and sub question number if any in the respective column provided in the answer sheet.
- 3) Do not write anything in the column provided for the marks to be assigned by the examiners.
- 4) Candidate will not be permitted to leave the examination hall until an hour after the question papers are distributed.
- 5) Every candidate present must sign against their seat number on the attendance sheet provided by the junior supervisor.
- 6) Candidates are forbidden to (i) bring any book, notes. Scribbling papers, mobile telephones, programmable calculators, or any other similar devices.(ii) speak or communicate in any manner to any other candidate, while the examination is in progress, and (iii) take with them any answer-book written or blank while leaving the examination hall. The supervisors/authorized persons are authorized to check the students.
- 7) Candidate suspected to be guilty of any of the aforesaid acts will be allowed to write their paper only after giving an undertaking in writing that the decision of the College Examination Authorities in respect of the reported act of unfair means is binding on them.
- 8) Candidates should write their answers legibly. They are warned that **zero marks** will be assigned to **answers which cannot be assessed by the examiners owing to illegible handwriting**.
- 9) Write on both side of a page. Rough work, when necessary, should be done on the Left- hand side and in pencil only.
- 10) While underlining of answers for focusing attention is permitted, use of varied inks, except for illustration and figures must be avoided.
- 11) No sheet shall be torn from the answer-books provided nor shall additional papers attached to them.
- 12) The answer-books will be scrutinized before they are sent to examiners.
- 13) All answer-books supplied shall be returned whether written or blank.
- 14) Nothing shall be written on the question-paper.
- 15) If candidate needs anything, they should approach the Junior Supervisor without disturbing other candidates. However, they should not leave their seat on my account.
- 16) A warning bell will be given ten minutes before the close of the examination. Candidates will not be allowed to leave the examination hall during the last ten minutes. At the final bell, they must stop writing and be ready to hand over their answer-books to the Junior Supervisor. They should not leave their seats until answers-books from all candidates are collected by the Junior Supervisor.
- 17) A candidate who disobeys any instructions issued by the Senior/Junior Supervisor or who is guilty of rude or disobedient behavior is liable for disciplinary action to be taken against him/her by the College Examination Authorities.



Space For Marks	Question No.	START WRITING HERE (Begin answer for each question on a new page)
	Q.1 a.	$f(w) = \frac{1}{2} \ xw - y\ ^2$ Gradient update Rule is .. $x_{t+1} = x_t - \eta \nabla f(x_t)$ Here $f(w) = \frac{1}{2} \ xw - y\ ^2$ $x_{w_{t+1}} = x_{w_t} - \eta \nabla \left(\frac{1}{2} \ xw - y\ ^2 \right)$ $\frac{\partial}{\partial w} = \frac{1}{2} \cdot 2 \cdot (xw - y) \cdot x$ $= x(xw - y)$ $= x^2 w - xy$ $w_{t+1} = w_t - \eta (x^2 w_t - xy)$ $= w_t - \eta x^2 w_t + \eta xy$ $= w_t (1 - \eta x^2) + \eta xy$ Convergence basically means how fast the x_t comes to local minima.

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(Begin answer for each question on a new page)

$$\frac{d}{dw^2} = xw^2 - xy$$

$$= 2xw$$

If the convergence needs to faster
so the value ~~is~~ lies between

$$0 < 2xw < 1$$

$$\frac{1}{2} \frac{ax^2 + by^2}{ax^2 + by^2}$$

or

and $\eta < \frac{2}{\lambda_{\max}}$

$$f_{wx} = 2xw - y$$

$$f_{wy} = -x$$

$$f_{yx} = -1$$

$$f_{xy} = -1$$

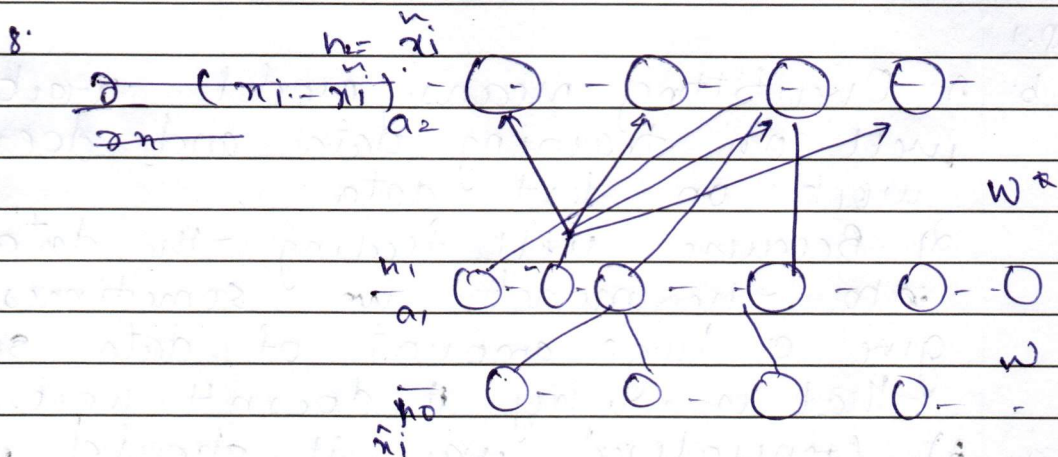
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	Q.2	
	b.	
		1) Under complete autoencoder is basically the no of features of a neuron in the hidden layer is more than the input. 2) $\dim(h) > \dim(x_i)$ $\uparrow$ $\uparrow$ hidden layer input layer 3) then what happen it will feedforward the x_i value to h and h will transmit to the $\hat{x}_i$. So it does not doing anything just copying the content via using hidden layer and transmit to output layer. 4) $\min_{w, w', p, r} \sum_{i=1}^n \sum_{j=1}^n (x_{ij} - \tilde{x}_{ij})^2$ i.e $\min_{w, w', p, r} \frac{1}{n} \sum_{i=1}^n (x_i - \tilde{x}_i)^T \cdot (x_i - \tilde{x}_i)$ 5) As MSE so we have divide by n. 6) PCA remove the features which are not that much required or it just combine the two features like wise how this under computer autoencoder do 7) Overcomplete encoder without regularization is not just transmitting the copying the content to h and then to $\hat{x}_i$.

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$$\frac{\partial L(\theta)}{\partial w} = \frac{\partial h_2}{\partial a_2} \cdot \frac{\partial a_2}{\partial h_1} \cdot \frac{\partial h_1}{\partial a_1} \cdot \frac{\partial a_1}{\partial w}$$

so the derivative of MSE is

$$\frac{d}{dx} = 2(x_i - \tilde{x}_i)$$

Now as in overcomplete encoder the x_i and $\tilde{x}_i$ is sample

$$= 2(x_i - x_i)$$

so we write this.

I (Identity)

$$I \cdot W_e \cdot W_d =$$

↑
weight
encoder

as $w_e \cdot w_d$ are same

I.

So hence proved that overcomplete autoencoder ~~ten~~ without regularization tends to learn the identity function.

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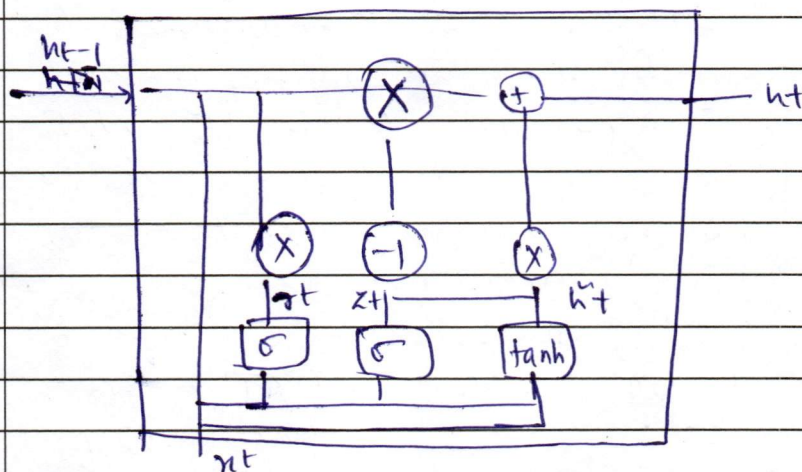
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Q.3
b.

1) GRU : Gated Recurrent Unit. It has two gates one is reset gate and update gate to make it simple from complexity.

2) Diagram of GRU:



$$r_t = \sigma(w_r \cdot [h_{t-1}, x_t]) \quad (\text{Reset Gate})$$

$$z_t = \sigma(w_z \cdot [h_{t-1}, x_t]) \quad (\text{Update Gate})$$

$$\tilde{h}_t = \tanh(w_h \cdot [h_{t-1}, x_t])$$

$$h_t = (1 - z_t) \cdot h_{t-1} + z_t \cdot \tilde{h}_t$$

Update gate is inputted as a convex combination of previous hidden state and candidate hidden state.

We calculate the candidate state that ~~for~~ taking the inputs as h_{t-1} and x_t and applying a tanh function that is $[-1, 1]$.

Putting into the hidden state taking into the consideration of previous update gate value & hidden state by adding with the current update gate value with hidden candidate state. This is how

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the update gate is interpreted of previous hidden state and candidate hidden state.

GRU has two gates that update gate and reset gate whereas LSTM has 3 gates input, forget and output.

so it reduces the complexity mathematically by 1 gate and in that gate the calculations is less. Mathematical Eqn..

LSTM :

$$f_t = \sigma(W_f[h_{t-1}, x_t] + b_f)$$

$$i_t = \sigma(W_i[h_{t-1}, x_t] + b_i)$$

~~$$o_t = \sigma(W_o[h_{t-1}, x_t] + b_o)$$~~

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c)$$

$$h_t = o_t \times \tanh(\tilde{c}_t)$$

Like some text is given "I am happy" so it help us to find the sentiment analysis in real word. we can find with LSTM also but GRU is faster and required less amount of computation.

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Q.3.

a.

$$\frac{\partial L}{\partial h_{t-k}} \text{ for RNN}$$

RNN Equation:

$$h_t = \tanh (w_h \cdot [h_{t-1}, x_t] + b_h)$$

- 1) Here what happens is the RNN model every time update its ~~upd~~ content like if it is important.
- 2) So what happens if the sentence ~~is~~ ~~con~~ has come before has vanished due to its updation.
- 3) So to overcome this limitation the LSTM come and it reverse three gate that is input, forget and output.
- 4) So like if the information ~~is~~ ~~not~~ it will not forget and if it is useless it forget that so the most important we can get from it as it was not getting when was RNN.
- 5) Vanishing gradients is some thing it vanishes the content ~~of~~ ~~for~~ which were their way before.

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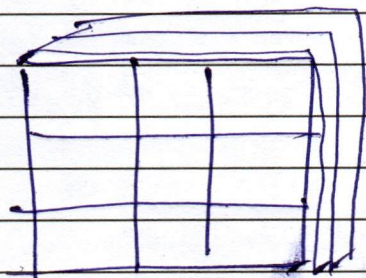
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Q.2

a.

2D convolution (number of multiplication) :-

Suppose 3×3



so the number of multiplication will be done is

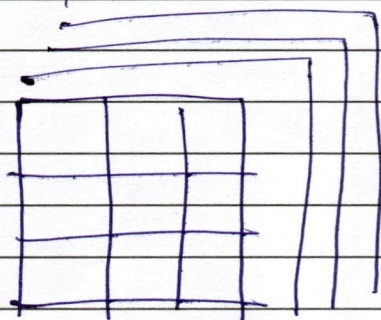
width $\times$ height $\times$ channels (R G B)

For the mention eg :- Red Green Blue

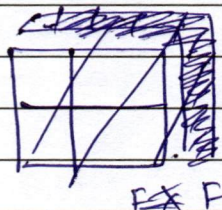
$$3 \times 3 \times 3 = 27$$

Depthwise Separable Convolution.

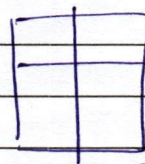
$H \times W \times C$ with K filters of size $F \times F$



$H \times W \times C$



$F \times F$

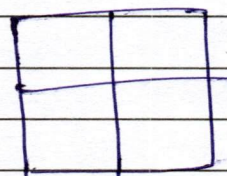


$F \times F$

$$= (H-1)(W-1)$$

will be

output of the image.



$(H-1)(W-1)$

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